

Set $\rightarrow$ collection of well defined & distinguished objects.
 $\hookrightarrow \{ \}$ Eg: collection of first 5 even natural no.'s.

Element of a set : $\{2, 4, 6, 8, 10\}$
 $\swarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \swarrow$
 Elements

Representation

Roaster $\{2, 4, 6, 8, 10\}$ listing all the elements	Set-Builder form $\{x \mid x \text{ is a prime number}\}$ specifying a common property of the elements
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Types of Set :-

a) Singleton set $\rightarrow$ set having only one element
 $\{1\}, \{\emptyset\}$

b) Finite set $\rightarrow$ set having finite number of elements
 Order of a finite set $\rightarrow$ number of elements it contains
 $A \rightarrow$ set $\quad \quad \quad O(A) \rightarrow$ order

c) Infinite set $\rightarrow$ containing infinite number of elements.
 Eg: set of natural numbers

d) Empty set $\rightarrow$ set containing no elements. ϕ
Eg: $\{u \mid u^2 = -2, u \in \mathbb{R}\}$

e) Universal set $\rightarrow$ set containing all possible values in the given frame of reference is called universal set. U OR E .

f) Subset (Improper Subset) $\rightarrow$ A and B two sets
Set A is said to be the subset of set B if each element of A is also an element of B.
 $\rightarrow$ Every set is a subset of itself
 $\rightarrow$ empty set is a subset of every set
 $\rightarrow$ Improper Subset $\rightarrow A \subseteq B$
A contains all elements of B
B contains all elements of A

g) Proper subset $\rightarrow$ if each element of A is in B and there is atleast one element in B which is not in A. i.e. $A \neq B$
denoted $\rightarrow A \subset B$

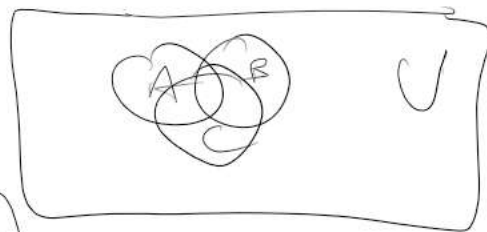
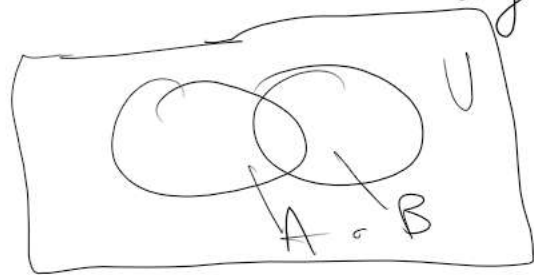
h) Equivalent sets $\rightarrow$ sets having same number of elements
 $n(A) = n(B)$.

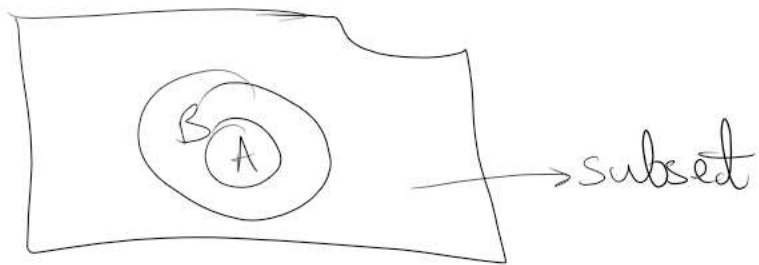
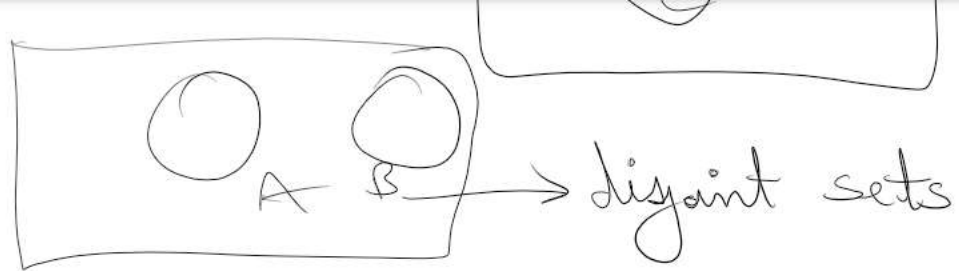
i) Equal sets $\rightarrow$ sets which have exactly same elements i.e. $A = B$
 $\rightarrow$ if A and B have to tell $A \subseteq B$ and $B \subseteq A$

j) Power set $\rightarrow$ set of all possible subsets of A
 $P(A) \rightarrow$ denoted

Venn Diagram

Venn - Euler diagram $\rightarrow$ geometrical representation of set.

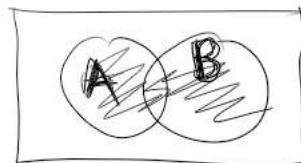




Operations on set

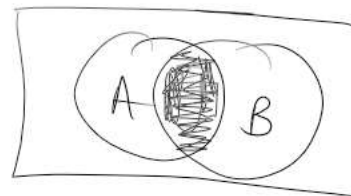
a) Union of set $(A \cup B)$

$$A \cup B = \{x: x \in A \text{ or } x \in B\}$$

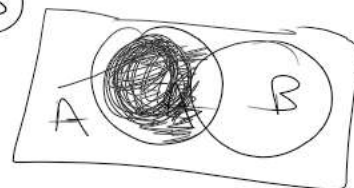


b) Intersection of set $(A \cap B)$

$$A \cap B = \{x: x \in A \text{ and } x \in B\}$$



c) Difference of set $(A - B)$
Symbolically:



$$A - B = \{x: x \in A \text{ and } x \notin B\}$$

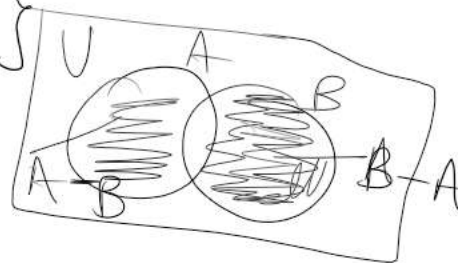
$$A - B = \{u : u \in A \text{ and } u \notin B\}$$

$$B - A = \{u : u \in B \text{ and } u \notin A\}$$

d) Symmetric difference of sets:

$$A \Delta B = \{u : u \in A - B \text{ or } u \in B - A\}$$

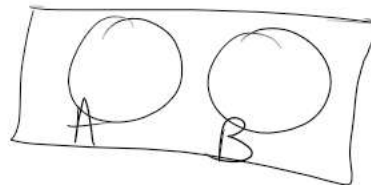
$$A \Delta B = (A \cup B) - (A \cap B) \\ (A - B) \cup (B - A)$$



Disjoint sets $\rightarrow$

no elements
 $A \cap B = \emptyset$

Common



Complement of sets:

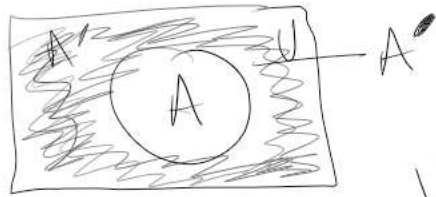
$A' \rightarrow$ denoted



$$A^C = \{u : u \in U \text{ and } u \notin A\}$$

Complement of sets:

$A' \rightarrow$ denoted



$$A^c = \{u : u \in U \text{ and } u \notin A\}$$

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 2, 3, 4\}$$

$$A' = \{5, 6, 7, 8\}$$

Laws of Algebra of sets:

a) Idempotent Law

$$i) A \cup A = A$$

$$ii) A \cap A = A$$

b) Identity Law

i)

$$A \cup \phi = A$$

$$ii) A \cap U = A$$

Identity elements for union and intersection

c)

Commutative Laws:

$$i) A \cup B = B \cup A$$

$$ii) A \cap B = B \cap A$$

c) Commutative Laws :
i) $A \cup B = B \cup A$
ii) $A \cap B = B \cap A$

d) Associative Law : A, B and $C \rightarrow$ sets

$$i) (A \cup B) \cup C = A \cup (B \cup C)$$

$$ii) A \cap (B \cap C) = (A \cap B) \cap C$$

union & intersection $\rightarrow$ associative.

e) Distributive Law :-

$$i) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$ii) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

f) De Morgan's Law :-

A and $B \rightarrow$ sets

$$i) (A \cup B)' = A' \cap B'$$

f) De Morgan's Law :-

A and B $\rightarrow$ sets

$$i) (A \cup B)' = A' \cap B'$$

$$ii) (A \cap B)' = A' \cup B'$$

#

Formula's
no. of elements in atleast one of the sets

$$i) n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$ii) n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

no. of elements in exactly one of the sets A, B & C is

$$= n(A) + n(B) + n(C) - 2n(A \cap B) - 2n(A \cap C) - 2n(B \cap C) + 3n(A \cap B \cap C)$$

no. of elements in exactly two of the sets A, B & C

$$= n(A \cap B) + n(B \cap C) + n(A \cap C) - 3n(A \cap B \cap C)$$

Ordered pair $\rightarrow (a, b)$
 $\downarrow$ 1st element $\downarrow$ 2nd element Entries in a specific order

Cartesian product / direct product of sets:

The set of all ordered pairs of elements $(a, b): a \in A$ and $b \in B$.

denoted $\rightarrow A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$

Relation

A & B $\rightarrow$ two non-empty sets. A relation from set A to set B is a subset of $A \times B$.

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$$A = \{1, 2, 3\}, \quad B = \{a, b\}$$

$$R_1 = \{(1, a), (2, a), (3, a)\}$$

$\hookrightarrow$ relation

$$R_2 = \{(1, a), (3, a), (4, b)\}$$

$\hookrightarrow$ not a relation

Eg: $A = \{1, 2\}$

$$B = \{1, 2, 3\}$$

R defined on $A \times B$

$$R = \{(a, b) : (a < b)\}$$

$$= \{(1, 2), (1, 3), (2, 3)\}$$

Domain and Range

If R is relation from A to B , then the set of first elements of elements of R is called the domain of R .

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If R is relation from A to B , then the set of first elements of elements of R is called the domain of R and the set of second elements of elements of R is called the range of R .

$$\text{Domain} \subseteq A$$

$$\text{Range} \subseteq B$$

No. of Relations :

$$A \rightarrow m$$

$$B \rightarrow n$$

No. of Relations : $A \rightarrow m$
 $B \rightarrow n$

$A \times B \rightarrow mn \rightarrow$ ordered pairs

So, total no. of subsets $\rightarrow 2^{mn}$
 $\downarrow$
no. of relations

Types of Relation:

$A \rightarrow$ Universal Relation $\rightarrow$ A relation R in a set A is called a universal relation in A if $R = A \times A$.

$\rightarrow$ Void relation = A relation R in a set A is called a void relation in A if $R = \phi$

Eg: $A = \{1, 2, 3\}$ $R \quad aRb = a-b = \sqrt{2}$

→ Void relation = A relation R in a set A is called a void relation in A if $R = \emptyset$

Eg: $A = \{1, 2, 3\}$ $R \ aRb = a - b = \sqrt{2}$

$$R = \emptyset \subseteq A \times A \rightarrow \text{void relation}$$

→ Identity relation = A relation R in a set A is called the identity relation A if $R = \{(a, a); a \in A\}$

$A = \{a, b, c\}$

identity relation $\{(a, a), (b, b)\}$

→ Reflexive relation → A relation R in a set A is called a reflexive relation if $(a, a) \in R \forall a \in A$.

→ Symmetric relation → A relation R in a set A is called a symmetric relation if $(a, b) \in R$ then $(b, a) \in R$
i.e. $\Rightarrow aRb \Rightarrow bRa$.

→ Antisymmetric relation → A relation R on a set A is called antisymmetric relation iff $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$ for all $a, b \in A$

NOTE - if $(a, b) \in R$ and $(b, a) \notin R$ then also R is an antisymmetric relation.

→ Transitive relation - a relation R on a set A is called a transitive relation if $(a, b), (b, c) \in R$ then $(a, c) \in R$ i.e. $aRb, bRc \Rightarrow aRc$.

→ Equivalence relation - A relation R on a set A is called an equivalence relation if R is reflexive, symm. and transitive. So, for an equivalence relation R in A .

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$$I) aRa \forall a \in A$$

$$II) aRb \Rightarrow bRa$$

$$III) aRb, bRc \Rightarrow aRc$$

→ Intersection of two equivalence relations on a set → equivalence
→ Union may not be.

→ $A, B \rightarrow$ two sets $R: A \rightarrow B$.
inverse of R denoted by R^{-1}
relation from $B \rightarrow A$.

Then the
and is a

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

→ Intersection of two equivalence relations on a set → equivalence
→ Union may not be.

→ $A, B \rightarrow$ two set $R: A \rightarrow B$. Then the
inverse of R denoted by R^{-1} and is a
relation from $B \rightarrow A$.

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

$$\text{Domain of } R = \text{Range of } R^{-1}$$

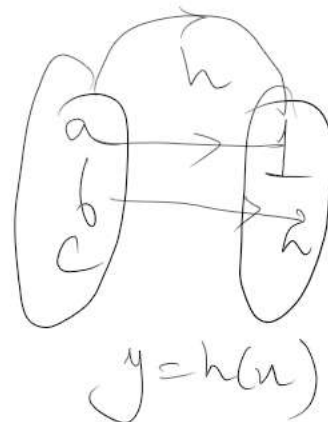
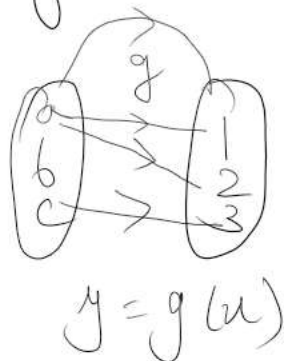
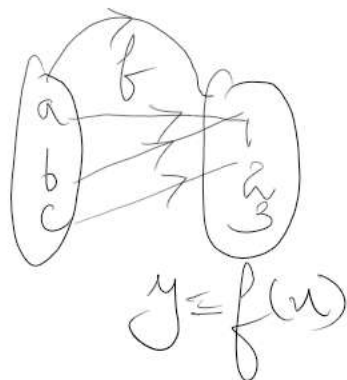
$$\text{Range of } R = \text{Domain of } R^{-1}$$

FUNCTIONS

function: $A \rightarrow B$ each and every element of Set A is associated
non-empty sets with a unique element of set B .
denoted by $f: A \rightarrow B$

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 non-empty sets
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here $g(x)$ and $h(x)$ are not functions
 $\downarrow$
 unique element
 $\rightarrow$ c does not have an image

Every function from $A \rightarrow B$ satisfies the following condition:

$\Rightarrow f \subset A \times B$

$\Rightarrow$

$\Rightarrow$

$\forall a \in A \Rightarrow (a, f(a)) \in f$
 $(a, b) \in f \text{ \& \& } (a, c) \in f \Rightarrow$

Every function from $A \rightarrow B$ satisfies the following condition:

$$i) f \subset A \times B$$

$$ii) \forall a \in A \Rightarrow (a, f(a)) \in f$$

iii)

$$(a, b) \in f \text{ \& } (a, c) \in f \Rightarrow b = c$$

Domain, Codomain and Range of a function

$$f: A \rightarrow B$$

domain

Elements of set B which are associated with set $A \rightarrow$ Range.

$$y = f(x)$$

$$\text{Domain} = \{a, b, c\}$$

$$\text{Codomain} = \{1, 2, 3\}$$

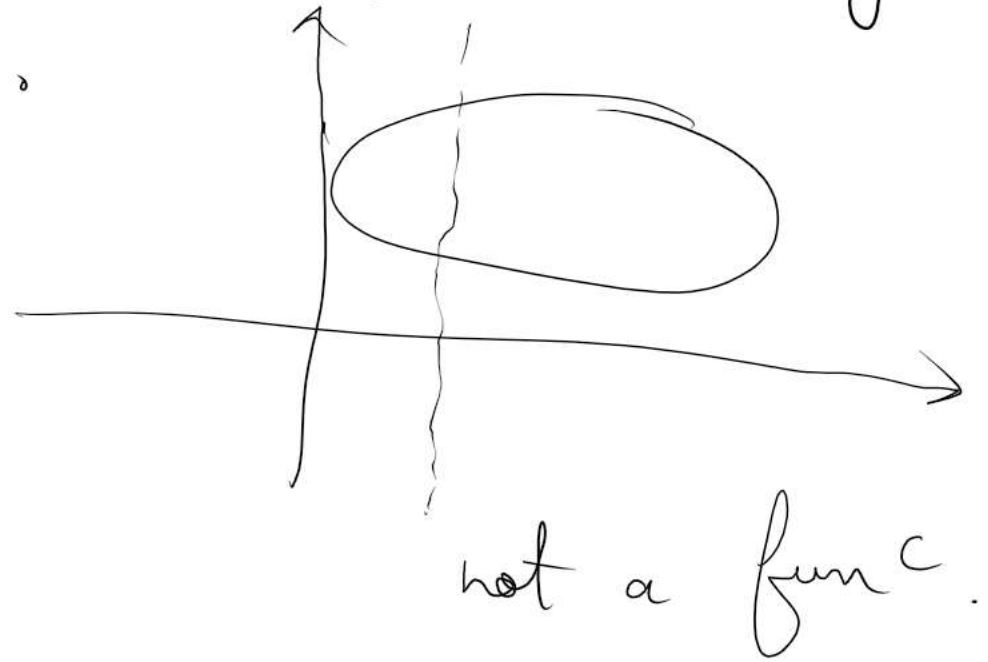
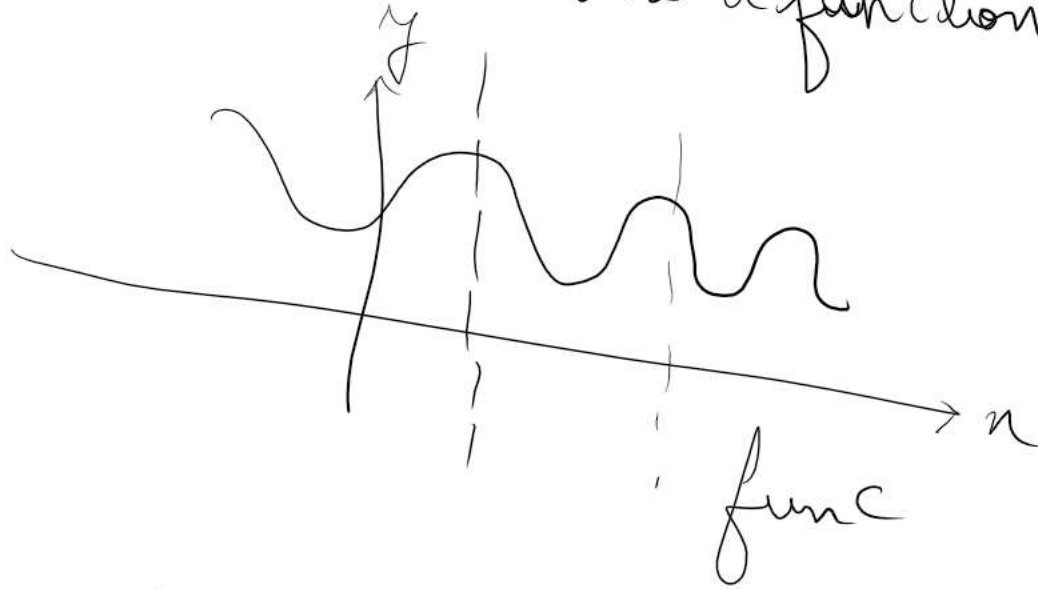
$$\text{Range} = \{1, 2\}$$

$$\text{Range} \subseteq \text{Codomain}$$

Range $\subseteq$ Codomain

Graphical Representation

In the graph of $y = f(x)$ if a straight line parallel to y-axis is drawn and if cut the graph at only one point then it is a function.



Classification of func^s:

Classification of functions:

a) Algebraic functions

i) Polynomial functions: $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$
an of degree n

$$D = \mathbb{R}$$

If $n \in \text{odd}$

$$\text{Range} = \mathbb{R}$$

If $n \in \text{even}$

→ Range depends on the given function

ii) Rational function

$$R(x) = \frac{P(x)}{Q(x)}, \quad Q(x) \neq 0$$

$$\text{It } D = \mathbb{R} - \{ \text{solutions of } Q(x) = 0 \}$$

$R =$ depends upon the function

b) Transcendental Functions:-

i) Trigonometric Functions:-

Function

Domain

Range

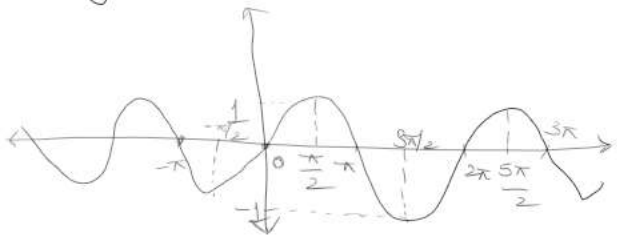
...
upon the function

6) Transcendental Functions:-

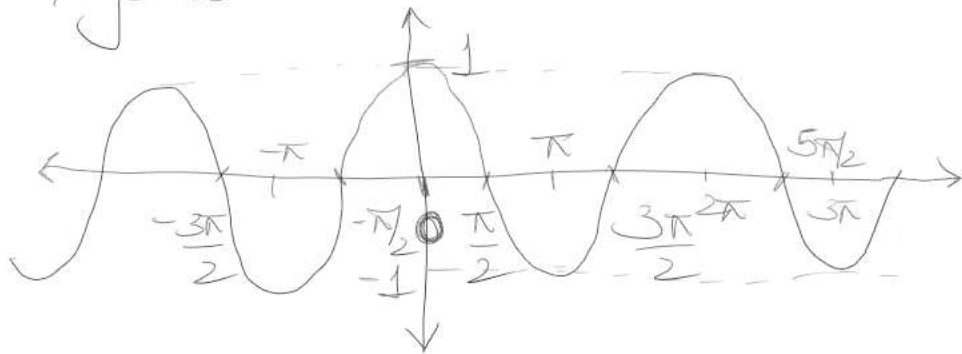
1) Trigonometric Functions:-

Function	Domain	Range
$\sin x$	$\mathbb{R}$	$[-1, 1]$
$\cos x$	$\mathbb{R}$	$[-1, 1]$
$\tan x$	$\mathbb{R} - \left\{ x : x = (2n+1)\frac{\pi}{2} \right\}$	$\mathbb{R}$
$\cot x$	$\mathbb{R} - \{ x : x = n\pi \}$	$\mathbb{R}$
$\sec x$	$\mathbb{R} - \left\{ x : x = (2n+1)\frac{\pi}{2} \right\}$	$\mathbb{R} - (-1, 1)$
$\operatorname{cosec} x$	$\mathbb{R} - \{ x : x = n\pi \}$	$\mathbb{R} - (-1, 1)$

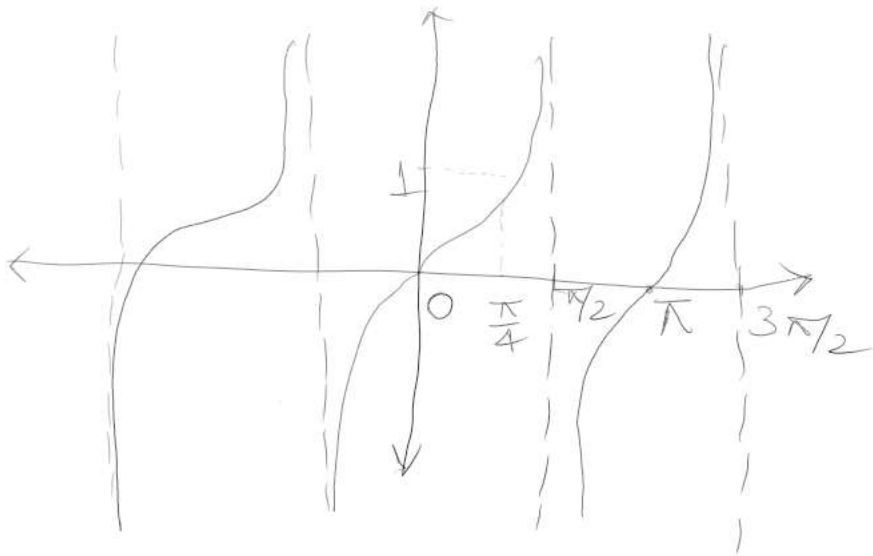
$$1) y = \sin u$$



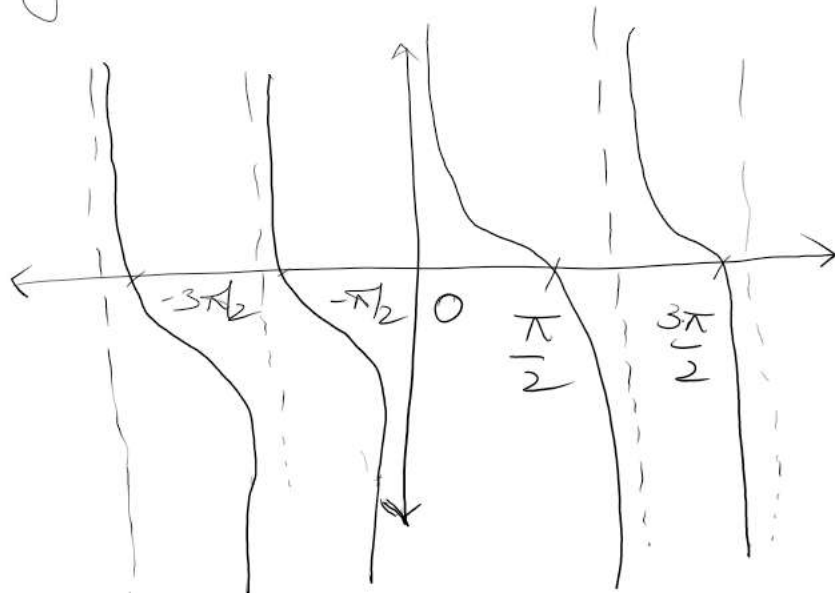
$$2) y = \cos u$$

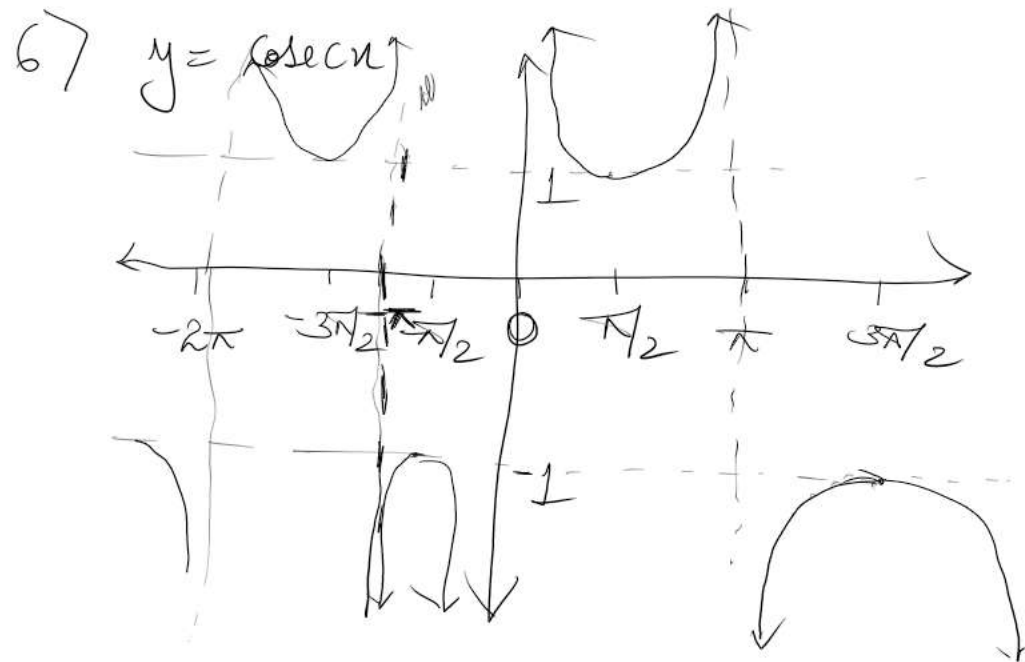
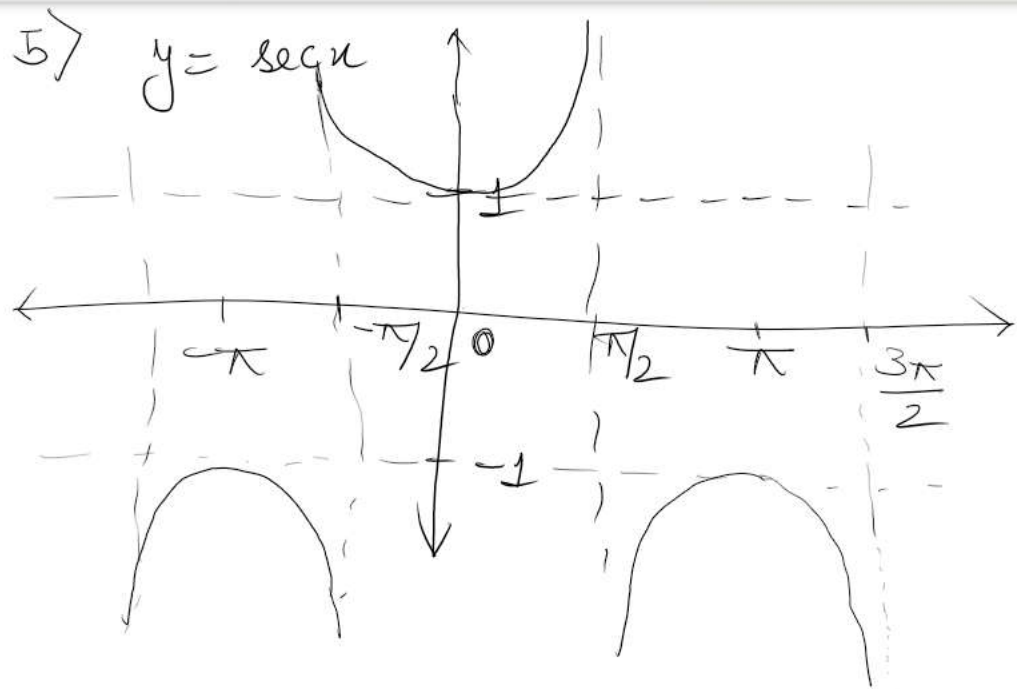


$$3) y = \tan u$$



$$4) y = \cot u$$





17 Inverse Trigonometric Functions

Function

Domain

Range

$$y = \sin^{-1} x$$

$$-1 \leq x \leq 1$$

$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$y = \cos^{-1} x$$

$$-1 \leq x \leq 1$$

$$[0, \pi]$$

$$y = \tan^{-1} x$$

$$-\infty < x < \infty$$

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$y = \cot^{-1} x$$

$$-\infty < x < \infty$$

$$(0, \pi)$$

$$y = \sec^{-1} x$$

$$x \leq -1 \text{ or } x \geq 1$$

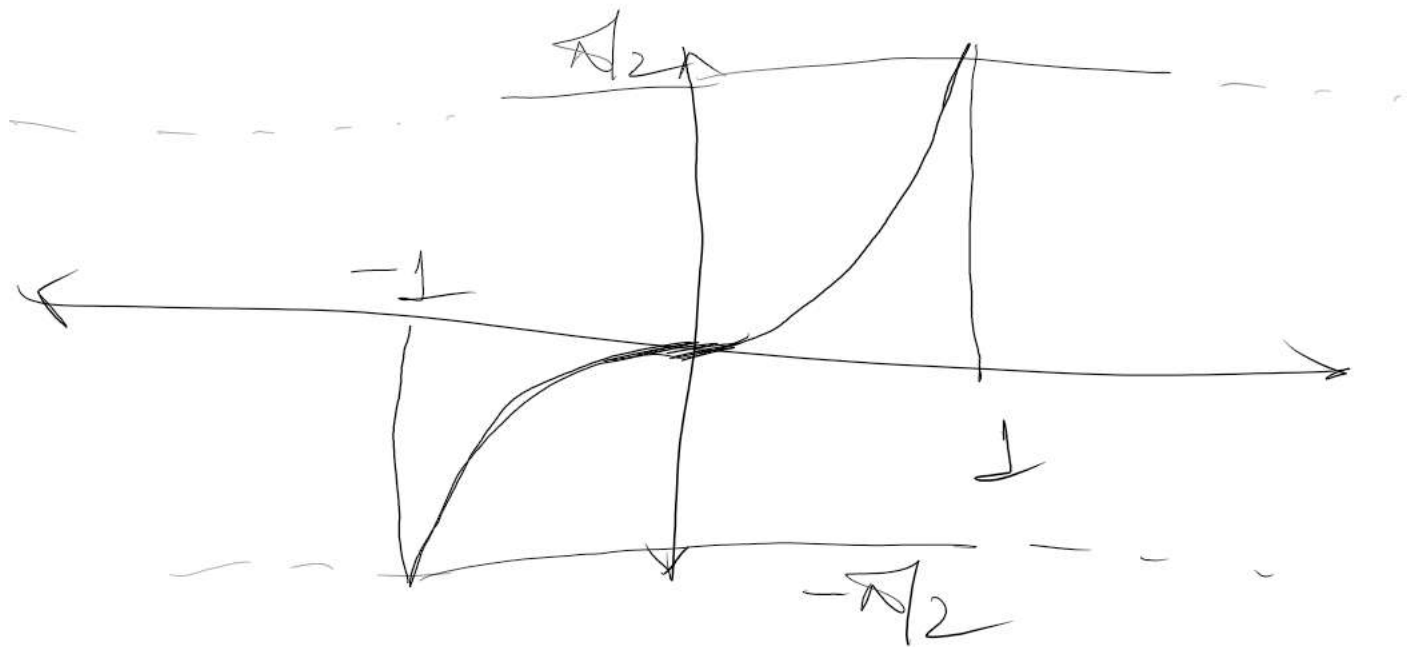
$$[0, \pi] - \left\{\frac{\pi}{2}\right\}$$

$$y = \operatorname{cosec}^{-1} x$$

$$x \leq -1 \text{ or } x \geq 1$$

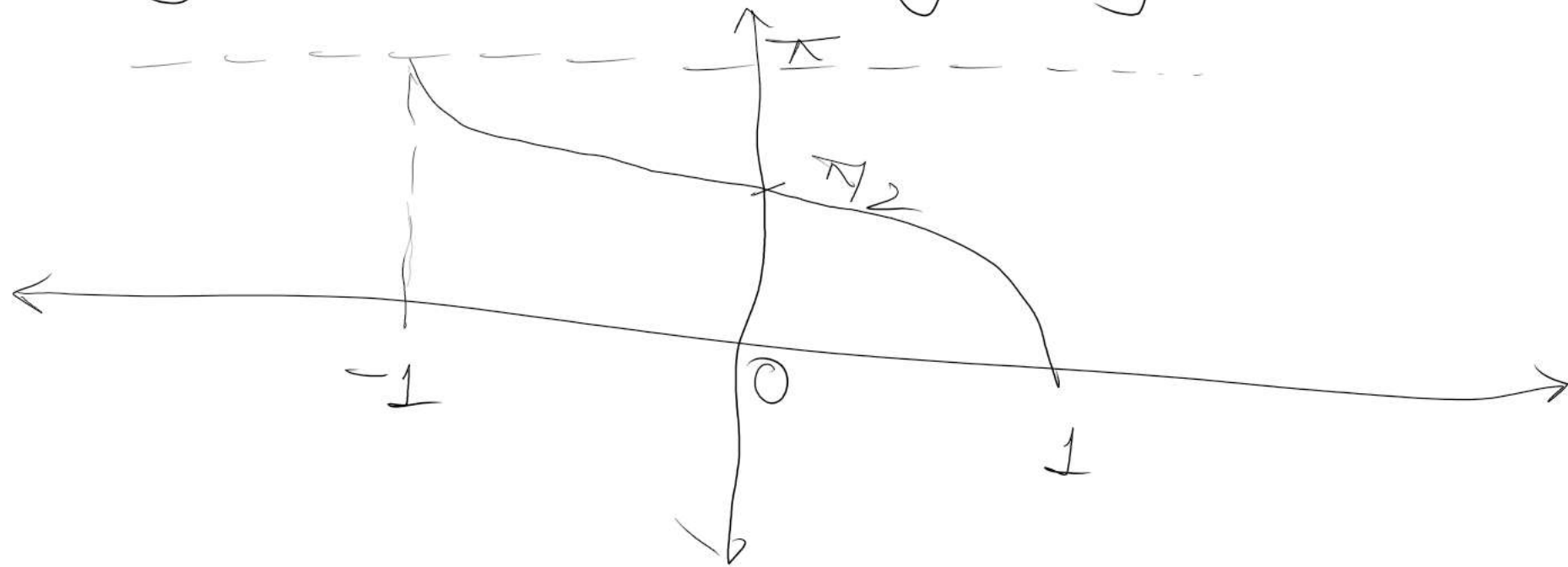
$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$$

17 $y = \sin^{-1} x$ $|x| \leq 1$, $y \in [-\pi/2, \pi/2]$

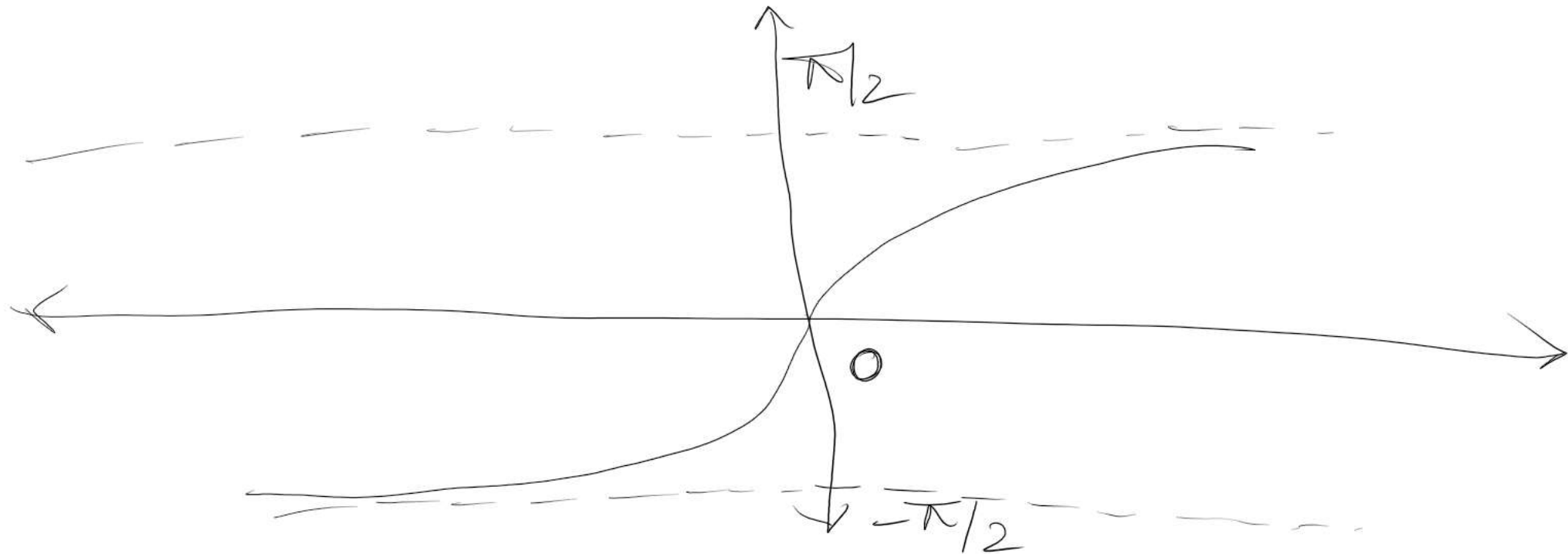


$= \pi/2$

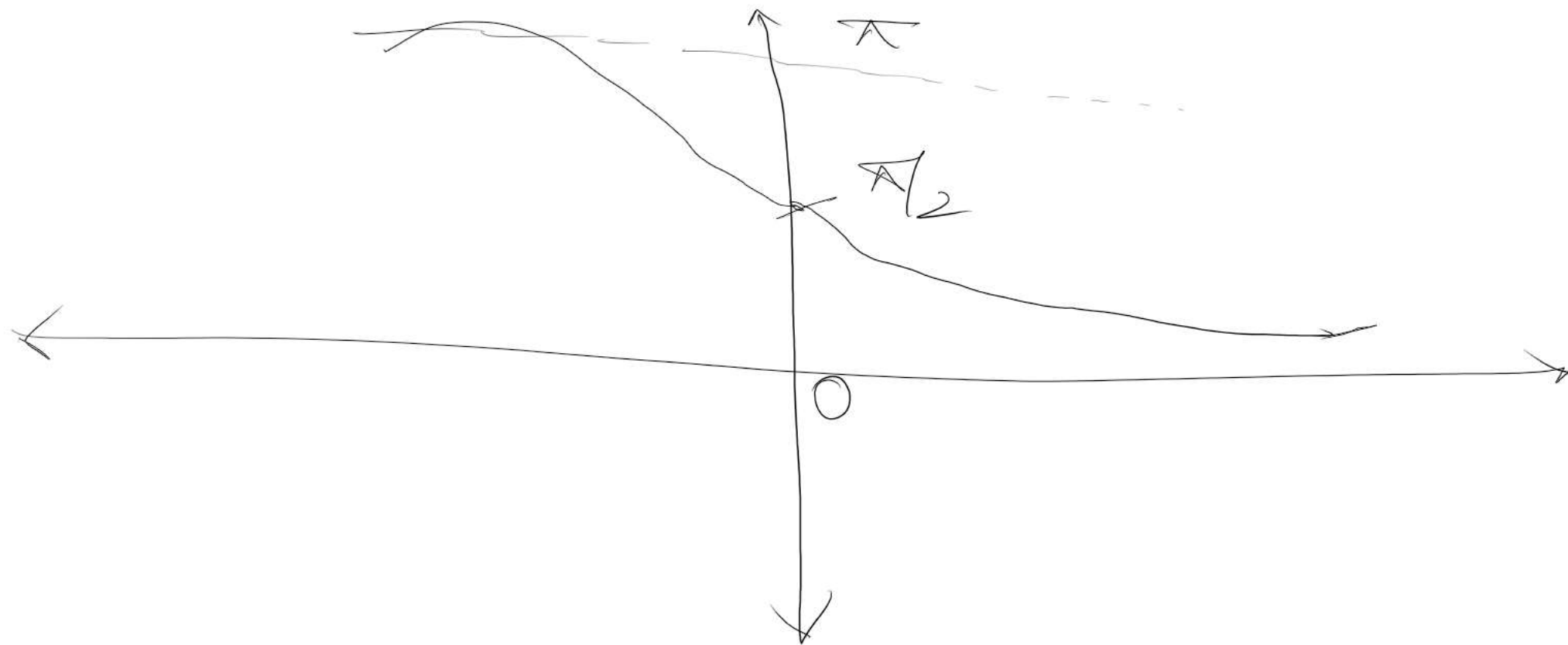
2) $y = \cos^{-1} x, |x| \leq 1, y \in [0, \pi]$



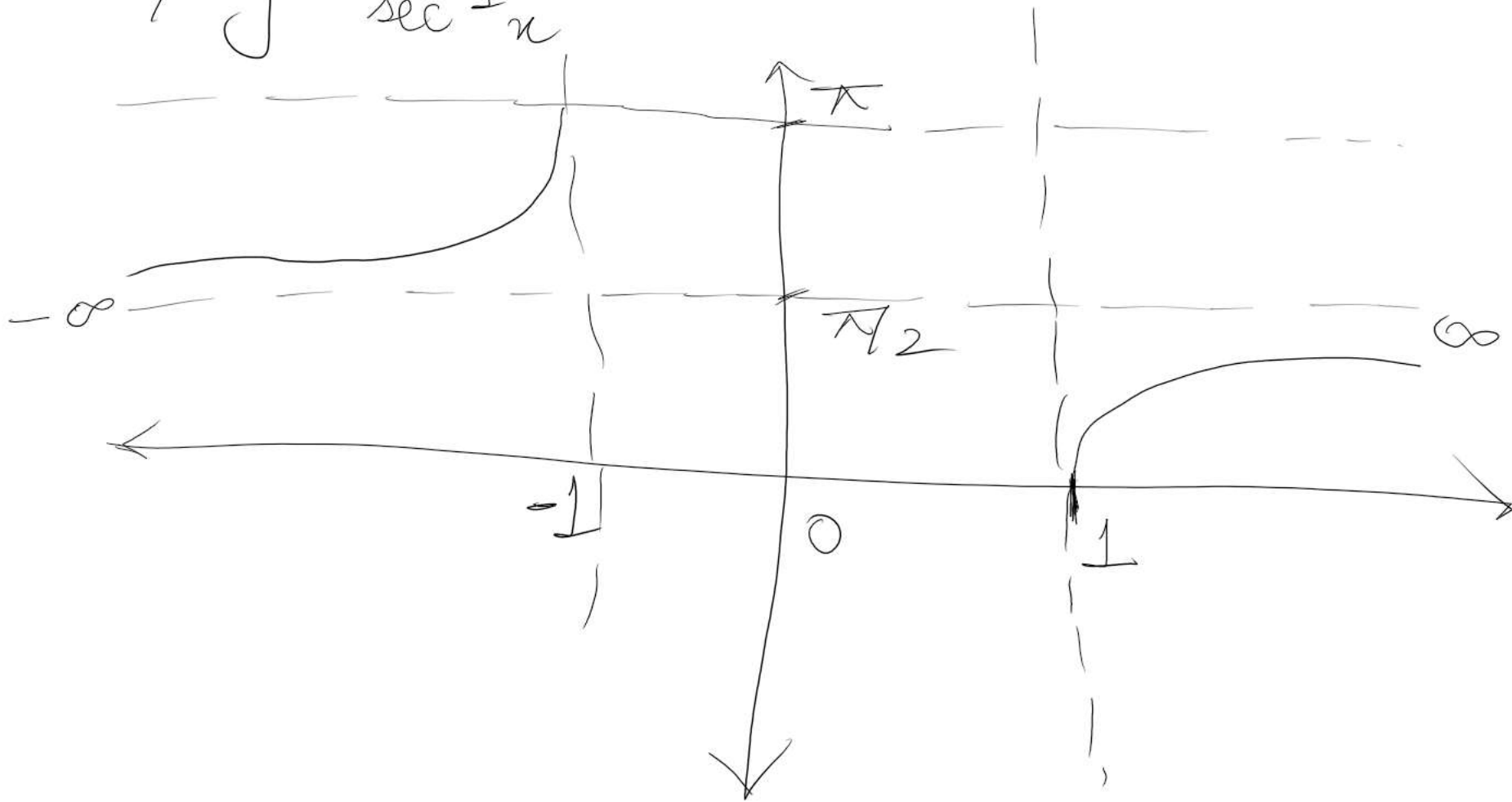
3) $y = \tan^{-1} x$



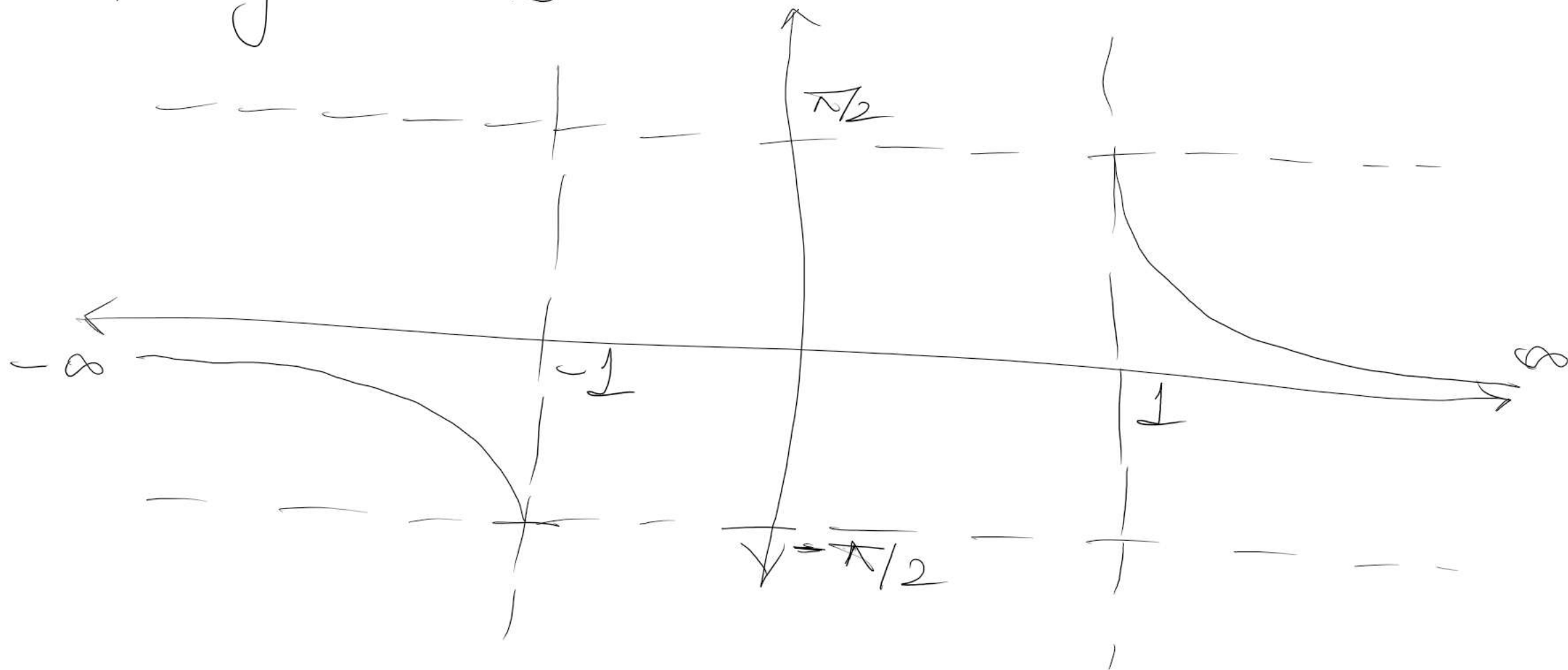
$$47 \quad y = d^{-1}n$$



$$\rightarrow y = \sec^{-1} x$$



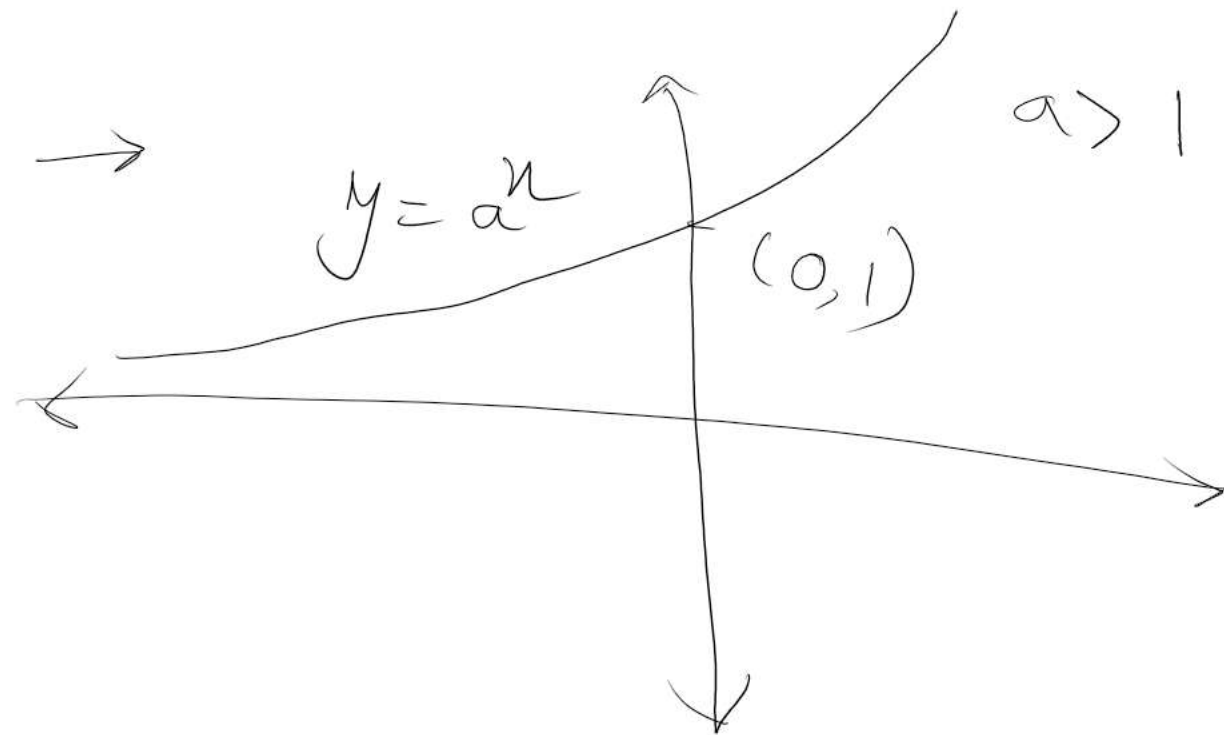
6) $y = \operatorname{cosec}^{-1} x$



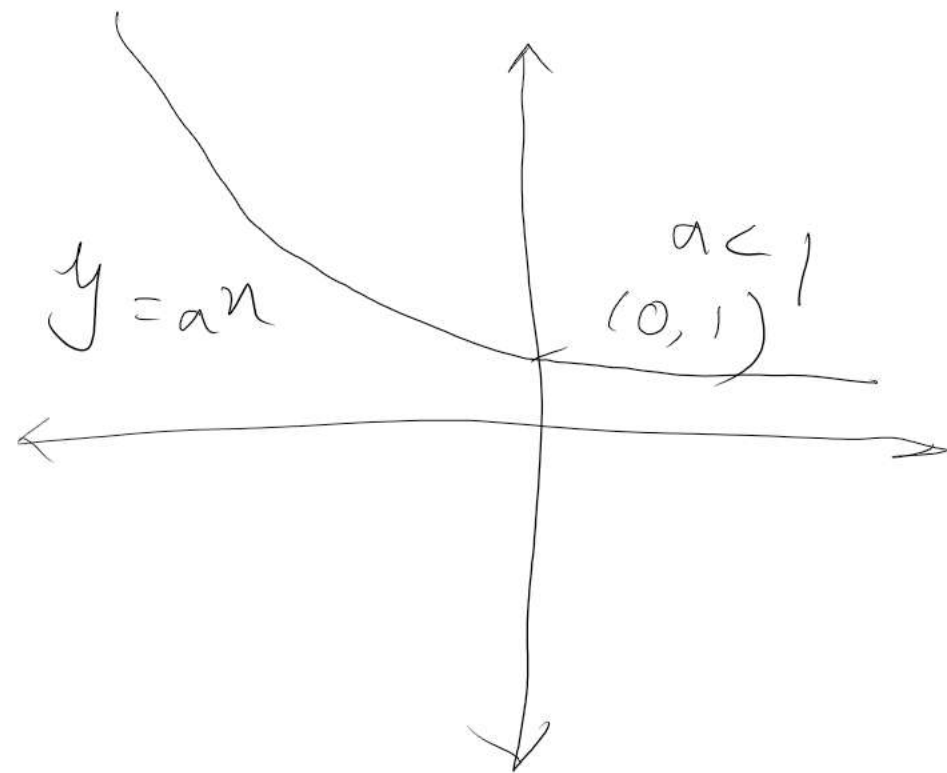
III > Exponential & Logarithmic Functions

a) $f(u) = a^u \rightarrow (a > 0, u \in \mathbb{R}) \rightarrow$ exponential function

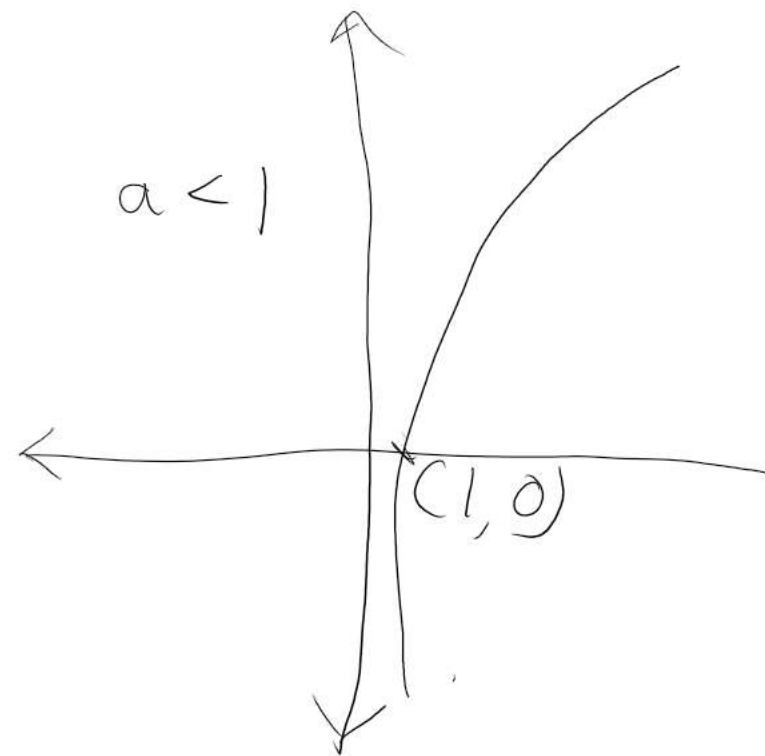
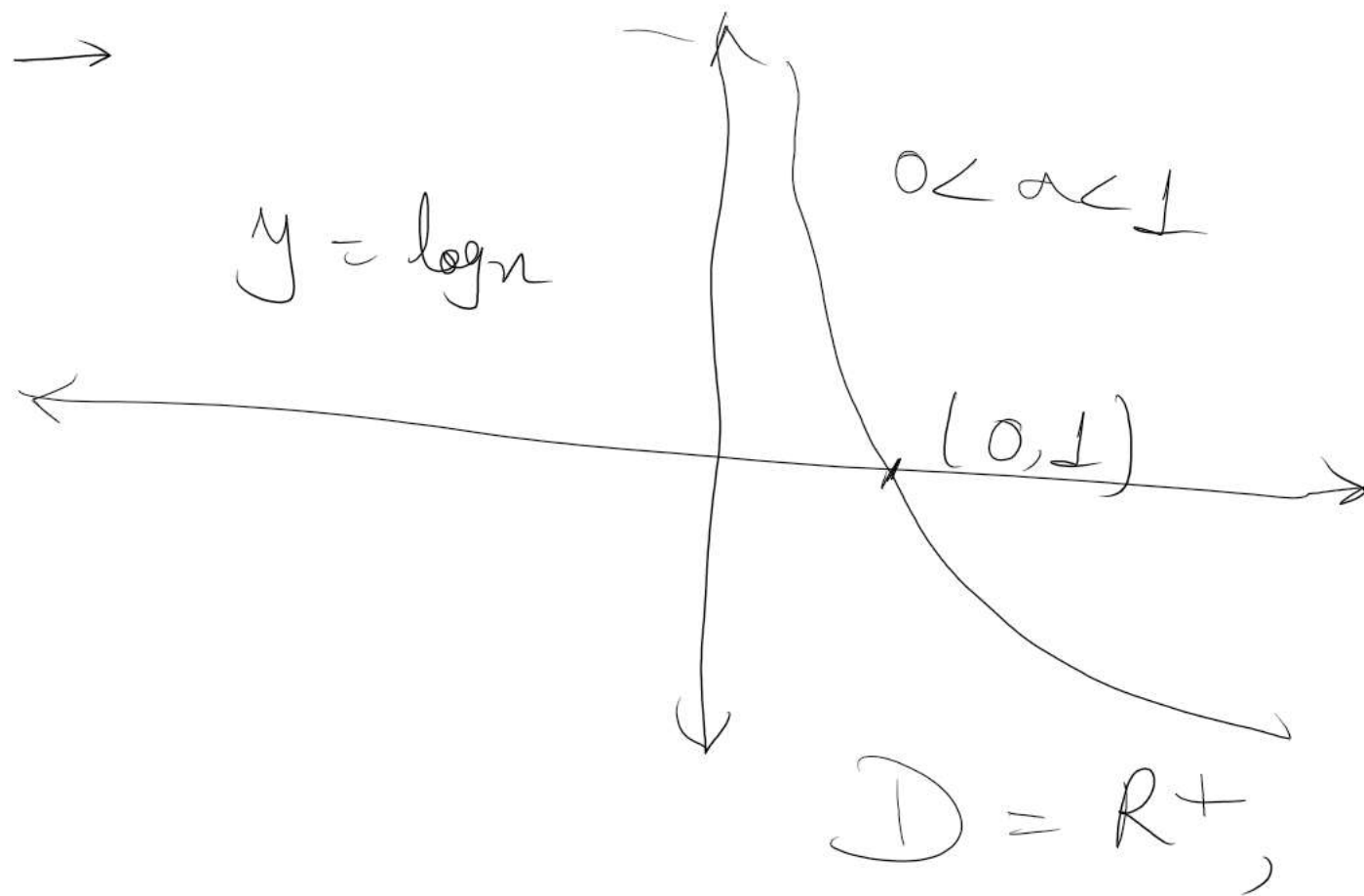
b) $g(u) = \log_a u \rightarrow$ logarithmic function



$D = \mathbb{R}$



$$D = \mathbb{R}, \quad \text{Range} = \mathbb{R}^+$$



c) Implicit and Explicit Functions

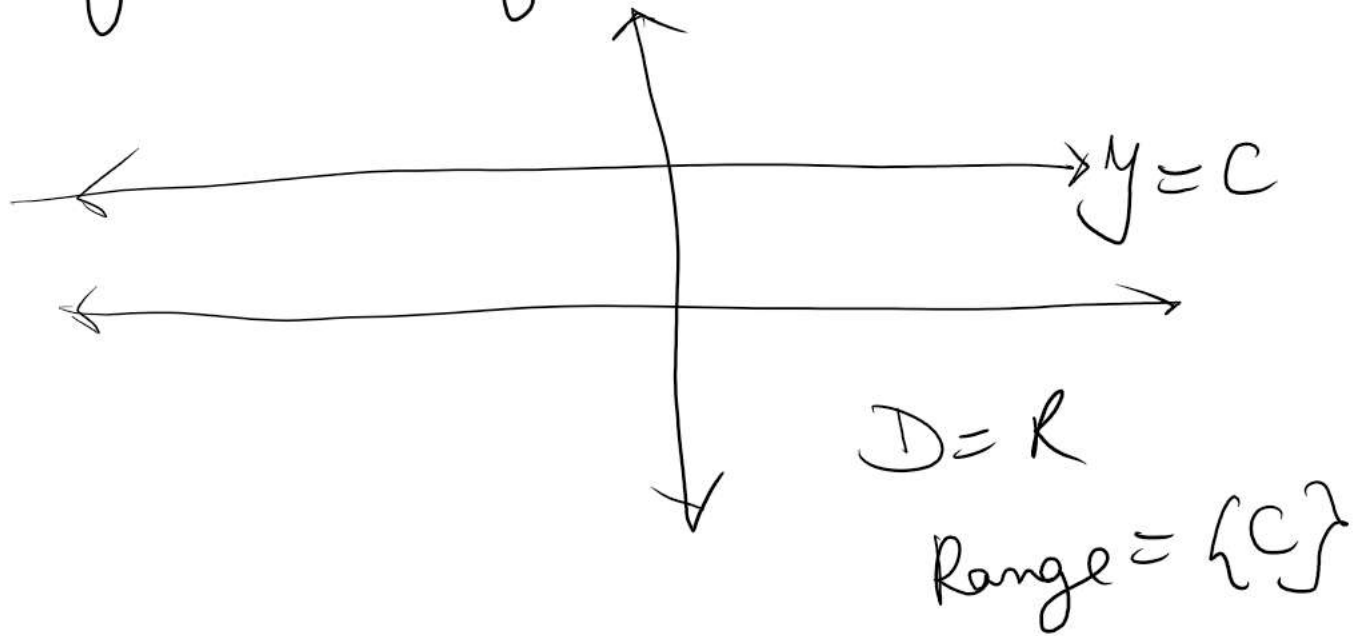
Implicit $\text{func} \rightarrow \text{func}$ where the relationship b/w variables is given without directly solving one variable in terms of another.

$$x^2 + y^2 = 25 \rightarrow \text{implicit}$$

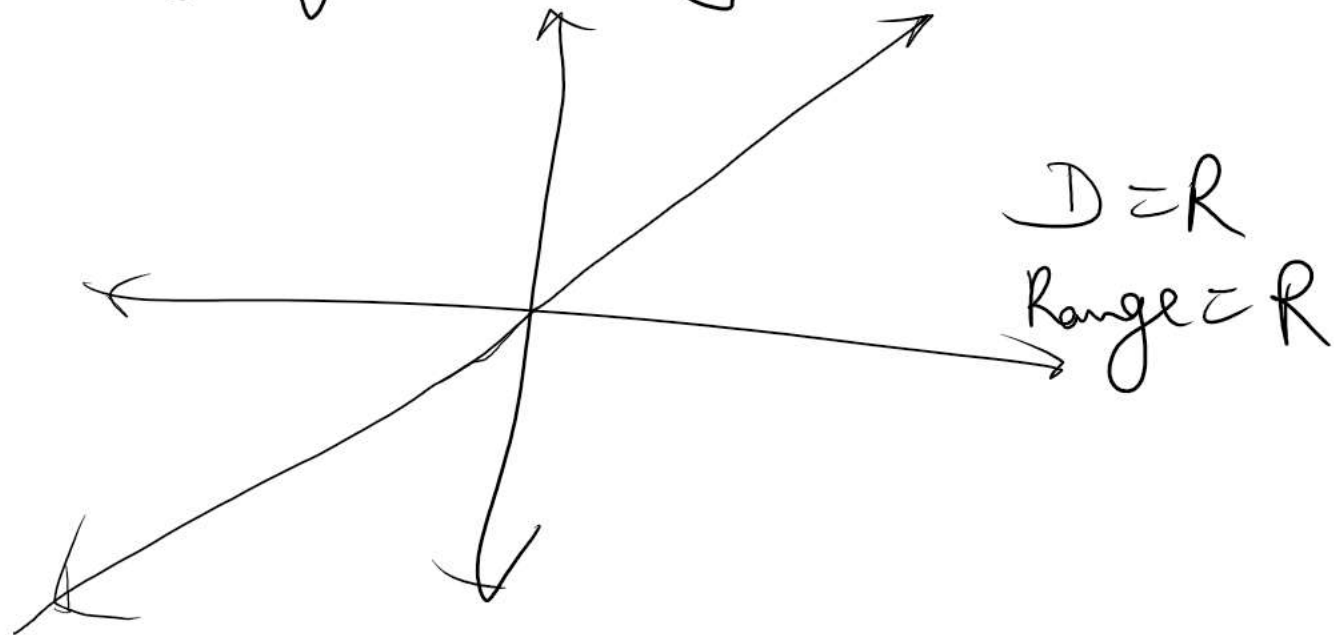
$$y = \pm \sqrt{25 - x^2} \rightarrow \text{explicit} \rightarrow y \text{ expressed in terms of } x.$$

Some other functions

1) Constant function : $f(x) = c$



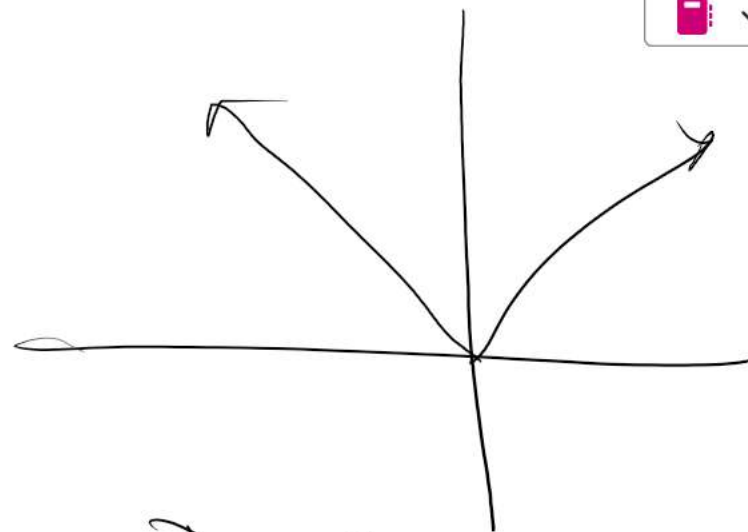
2) Identity function $y = x$



3) Modulus function $f(x) = |x|$

$$|x| = x \quad x \geq 0$$

$$= -x \quad x < 0$$



$$D = \mathbb{R}$$

$$R = [0, \infty)$$

4) Signum function

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$$\operatorname{sgn} u = 1 \quad u > 0$$

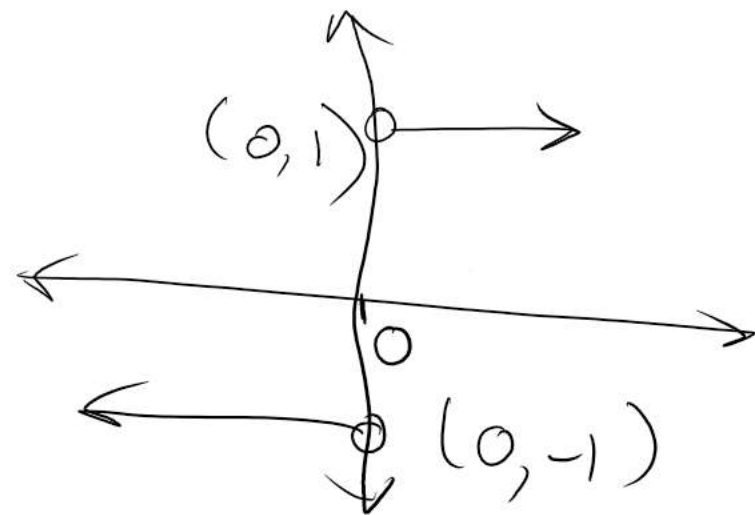
$$0 \quad u = 0$$

$$-1 \quad u < 0$$

$$\operatorname{sgn}(u) = |u|/u$$

$$0$$

$$\mathbb{R} = [0, \infty)$$



$$u \neq 0 \quad \mathbb{D} = \mathbb{R}$$

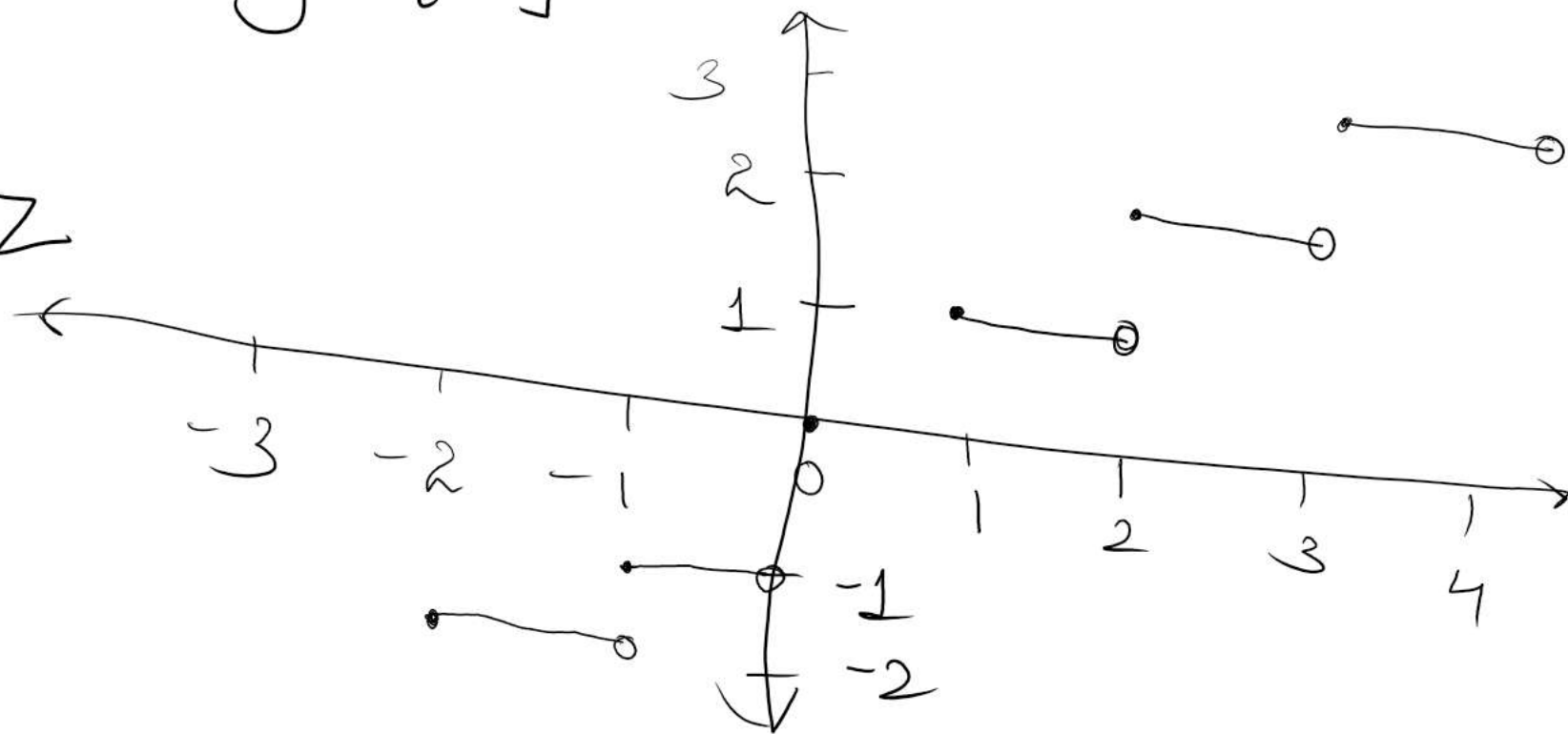
$$\mathbb{R} = \{1, 0, -1\}$$

$$u = 0$$

5) Greatest Integer Function / Step func :-
 $y = [x]$

$$D = \mathbb{R}$$

$$\text{Range} = \mathbb{Z}$$



Properties :

$$I) \quad [n] \leq n < [n] + 1$$

$$II) \quad n - 1 < [n] \leq n$$

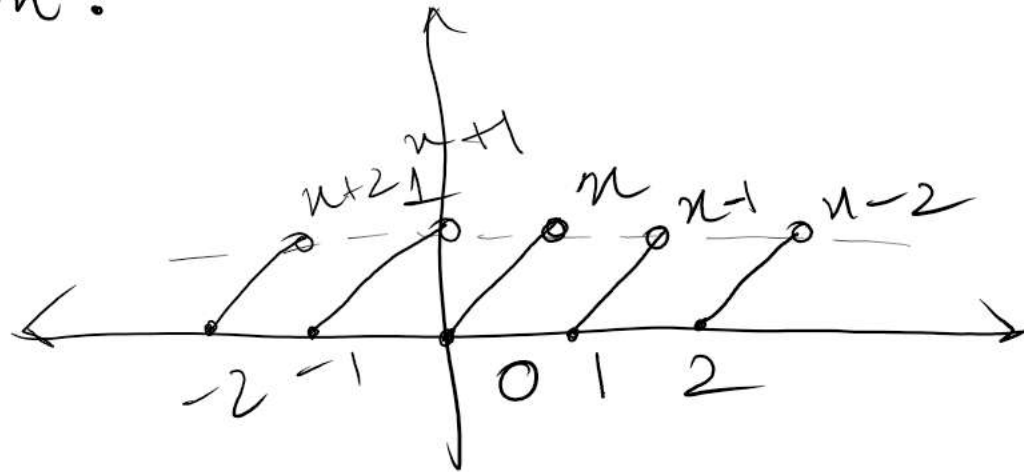
$$III) \quad [n+m] = [n] + m \rightarrow m \text{ is an integer}$$

$$IV) \quad [n] + [-n] = 0 \quad n \in \mathbb{Z} \\ = -1 \quad n \notin \mathbb{Z}$$

6) Fractional part Function:

$$0 \leq x < 1 \quad \{x\} = x - [x]$$
$$\{x\} = x - 0$$

$$1 \leq x < 2$$
$$\{x\} = x - 1$$



DOMAIN

RANGE

$$f(x) = \sqrt{x^2 - 5x + 6}$$

$$f(x) = \sqrt{1-2x} + 3\sin^{-1} \frac{3x-1}{2}$$

$$f(x) = \sqrt{\log_{1/2} \left(\frac{5x - x^2}{4} \right)}$$

$$f(x) = \sqrt{\frac{x-2}{x+2}} + \sqrt{\frac{1-x}{1+x}}$$

$$f(x) = \frac{3}{2-x^2}$$

$$y = \frac{x^2}{1+x^2}$$

$$y = \log_e (3x^2 - 4x + 5)$$

Equal Functions

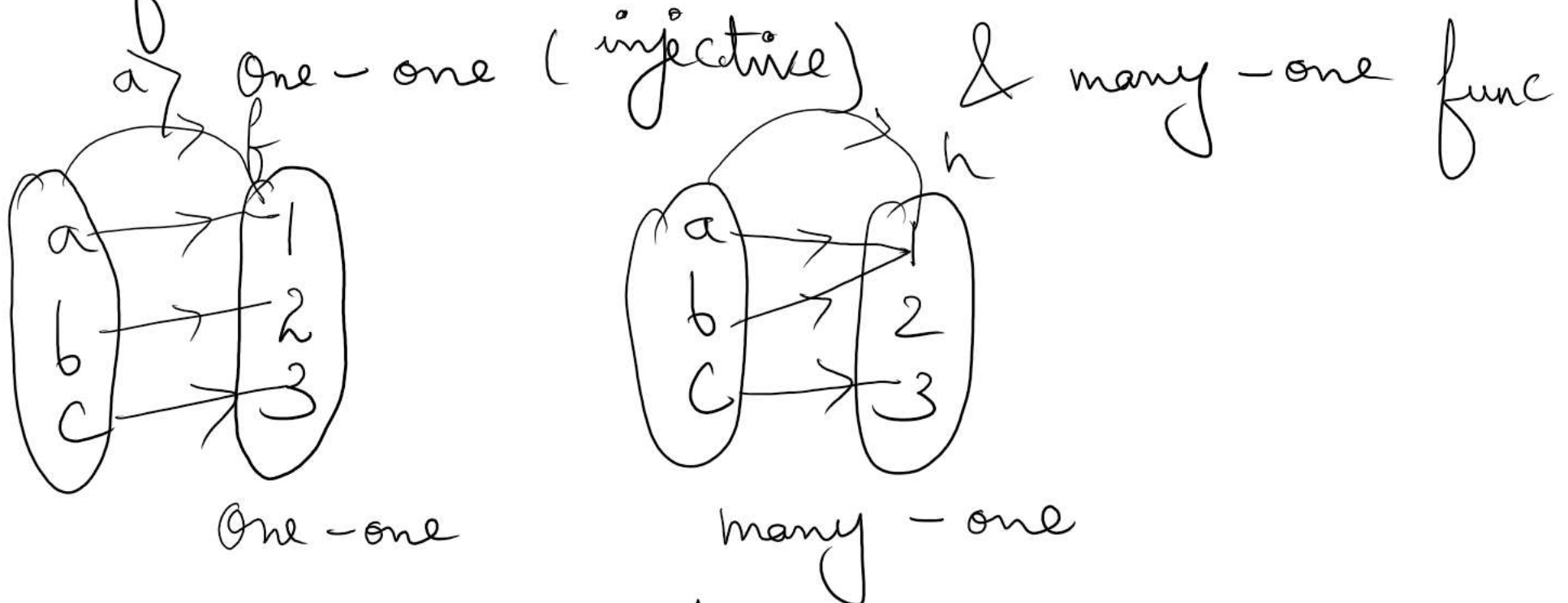
$f, g \rightarrow 2$ func's

I $D(f) = D(g)$

II $R(f) = R(g)$

III $\forall u \in D \quad f(u) = g(u)$

Types of Function :-



b) Onto & into function

$\downarrow$
Codomain = Range

if Codomain $\neq$ Range

c) Bijective function

A funcⁿ is called Bijective



```
graph TD; A[Bijective] -- both --> B[One-one]; A -- both --> C[Onto]
```

One-one Onto

Q. How to check one - one func?

i) $f(u_1) = f(u_2) \Rightarrow u_1 = u_2$

ii) Monotonicity $f'(u)$

iii) draw a line parallel to x-axis
if it cuts the curve only once
 $\rightarrow$ one - one func

Q

Check onto func :-

→ find any element in the codomain

which has no pre-image in
Set A then $f: A \rightarrow B$ will be
an into function

No. of Functions

Set $A \rightarrow m$ distinct elements

Set $B \rightarrow n$ distinct elements

$$f: A \rightarrow B$$

$n^m \rightarrow$ diff^x func's

${}^n P_m \rightarrow$ one-one func

Inverse Functions

$f: A \rightarrow B$ be a one-one & onto func, then
there exists a unique func!

$$g: B \rightarrow A \rightarrow f(x) = y \Leftrightarrow g(y) = x$$

$$\forall x \in A \text{ and } y \in B$$

then g is said to be inverse
of f .

then g is said to be inverse
of f .

$$g = f^{-1} : B \rightarrow A = \{ (f(u), u) \mid (u, f(u)) \in f \}$$

a) Properties of Inverse Function

→ Inverse func of a bijection is unique

→ $f : A \rightarrow B$ is a bijection & $g : B \rightarrow A$

$$f \circ g = I_B$$

$$g \circ f = I_A$$

Identity func of A and B

$$f \circ g = I_B$$

$$g \circ f = I_A$$

$\rangle$ Identity func of A and B respectively

inverse of f

III $\rangle$ graphs are mirror images of each other in the line $y=x$.

IV $\rangle$ Inverse func is also bijective

V $\rangle$ $f, g \rightarrow$ bijections then the inverse of $g \circ f$ exists and

$$f: A \rightarrow B, g: B \rightarrow C$$

✓> $f, g \rightarrow$ bijections then the inverse of $g \circ f$ exists and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$
 $f: A \rightarrow B, g: B \rightarrow C$

✓1> If graph of $f(x)$ and $f^{-1}(x)$ cut each other at least one of their points of intersection lies on $y=x$, so $f(x) = f^{-1}(x) \Rightarrow f(x) = x$

b) How to find inverse of a function
1> Let assume given func equal to y .

b) How to find inverse of a function

i) Let assume given func equal to y .

ii) write x in terms of y .

iii) Replace x by $f^{-1}(x)$ and y by x .

$$Q. f: \mathbb{R} \rightarrow \mathbb{R} \\ f(x) = 2x + 3, \quad f^{-1}(x)$$

Even and Odd Functions

a) Definition:-

for $x \in D$, if $-x \in D$ then

i) $f(x) \rightarrow$ even func

Even and Odd Functions

a) Definition:-

for $x \in D$, if $-x \in D$ then

$\Rightarrow f(x) \rightarrow$ even func
if $f(x) = f(-x)$

$\Rightarrow f(x) \rightarrow$ odd func
if $f(x) = -f(-x)$

b) Geometrical Property:-

Even func $\rightarrow$

Sym about y-axis

Odd func $\rightarrow$ Sym about origin

b) Geometrical Property :-
Even func $\rightarrow$ Symm about y-axis
Odd func $\rightarrow$ Symm about origin

c) Every func can be expressed as a sum of even and odd func

$$f(u) = \underbrace{\frac{f(u) + f(-u)}{2}}_{\text{Even part}} + \underbrace{\frac{f(u) - f(-u)}{2}}_{\text{Odd part}}$$

d) $f(n) = 0 \rightarrow$ both odd and even.

e) f and $g \rightarrow$ both even/odd
 $f \circ g \rightarrow$ even
 $\rightarrow$ if one is odd $f \circ g \rightarrow$ odd

~~##~~ . Composite Function

$$f: A \rightarrow B, \quad g: B \rightarrow C$$

$$g \circ f: A \rightarrow C$$

$$(g \circ f)(n) = g(f(n)) \quad \forall n \in A$$

$$g \circ f: A \rightarrow C$$

$$(g \circ f)(u) = g(f(u)) \quad \forall u \in A$$

is called composite of two functions

$$g \circ f \rightarrow f \circ g$$

the range of f must be a subset of domain g .

$$\text{Range of } f \subseteq \text{Domain of } g$$

Properties of Composite fun:-

$$1) \quad g \circ f \neq f \circ g$$

Properties of Composite func:-

$$I) g \circ f \neq f \circ g$$

$$II) (f \circ g) \circ h = f \circ (g \circ h)$$

Periodic Func's

$$f(T+n) = f(n)$$

$T \in \mathbb{R} \rightarrow$ least positive value called the fundamental period of function

1) Period of $\sin^n x$, $\cos^n x$, $\operatorname{cosec}^n x$, $\sec^n x$ is 2π
 $\pi \rightarrow n(\text{even})$ $(n \text{ odd})$

ii) Period of $\tan^n x$, $\cot^n x$ (n odd or even),
 $|\sin x|$, $|\cos x|$, $|\tan x|$, $|\cot x|$,
 $|\sec x|$, $|\operatorname{cosec} x| \rightarrow \pi$

iv) Constant func $\rightarrow$ period is indeterminate.

vi) Period of $f(x) \rightarrow T$

i) Period of $f(ax+b) = \frac{T}{|a|}$

1) Period of $f(an+b) = \frac{T}{|a|}$

11) $\sqrt{f(n)}, af(n), 1/f(n) \rightarrow T$

$f_1(n), f_2(n), f_3(n)$ are periodic
with T_1, T_2 and T_3

$$f_1(n) \pm f_2(n) \pm f_3(n) \rightarrow \text{LCM of } T_1, T_2 \text{ and } T_3$$

if they are even $\rightarrow \frac{1}{2}$ LCM of T_1, T_2
and T_3

d) LCM of fractions $\frac{a}{b}$ & $\frac{c}{d} = \frac{\text{LCM of } a \& c}{\text{HCF of } b \& d}$

i) LCM of a rational and an irrational number
for eg 1 and π does not exist

ii) LCM of two irrational no. of different kind
 $2\sqrt{3}$, $3\sqrt{2}$ does not exist

iii) Period of $f(g(x)) \rightarrow$ will be period of $g(x)$.